

## # Exercise 18.2

- Find, from the first principles, the differential coefficient of

①  $\sin 2x$

→ let  $f(x) = \sin 2x$

$$f(x+h) = \sin 2(x+h) \\ = \sin (2x+2h)$$

Where  $h$  is small change in  $x$ .

By definition of derivatives

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin (2x+2h) - \sin 2x}{h}$$

$$= \lim_{h \rightarrow 0} 2 \sin \left( \frac{2x+2h-2x}{2} \right) \cdot \cos \left( \frac{2x+2h+2x}{2} \right) \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} 2 \frac{\sin h}{h} \times \cos (2x+h)$$

$$= \lim_{h \rightarrow 0} 2 \times 1 \times \lim_{h \rightarrow 0} \cos (2x+h)$$

$$= 2 \times \cos (2x+0)$$

$$= 2 \times \cos 2x$$

$$= 2 \cos 2x$$



② b)  $\cos(ax+b)$

→ let  $f(x) = \cos(ax+b)$

$$f(x+h) = \cos[a(x+h)+b]$$

$$= \cos(ax+ah+b)$$

Where  $h$  is small change in  $x$ .

By definition of derivatives

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos(ax+ah+b) - \cos(ax+b)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ 2 \sin\left(\frac{ax+ah+b+ax+b}{2}\right) \cdot \sin\left(\frac{ax+b-ah-ax-b}{2}\right) \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ 2 \sin\left(\frac{2ax+2b+ah}{2}\right) \cdot -\sin\left(\frac{ah}{2}\right) \right]$$

$$= \lim_{h \rightarrow 0} -2 \sin\left(\frac{2ax+2b+ah}{2}\right) \cdot \lim_{h \rightarrow 0} \frac{\sin\frac{ah}{2}}{\frac{ah}{2}} \times \frac{a}{2}$$

$$= \lim_{h \rightarrow 0} -2 \sin\left(\frac{2ax+2b+ah}{2}\right) \cdot 1 \times \frac{a}{2}$$

$$= -\sin\left(\frac{2ax+2b+0}{2}\right) \cdot a$$

$$= -a \sin\left(\frac{2ax+2b}{2}\right)$$

$$= -a \sin(ax+b)$$



3) c)  $\tan(3x-4)$

→ let  $y = \tan(3x-4) = f(x)$

$f(x+h) = \tan[3(x+h)-4]$

Where  $h$  is small change in  $x$ .

By definition of derivatives.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\tan(3x+3h-4) - \tan(3x-4)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\sin(3x+3h-4)}{\cos(3x+3h-4)} - \frac{\sin(3x-4)}{\cos(3x-4)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\sin(3x+3h-4) \cdot \cos(3x-4) - \cos(3x+3h-4) \cdot \sin(3x-4)}{\cos(3x+3h-4) \cdot \cos(3x-4)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\sin(3x+3h-4) - 3x - \cancel{3x} + \cancel{3x} - 4}{\cos(3x+3h-4) \cdot \cos(3x-4)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\sin 3h}{\cos(3x+3h-4) \cdot \cos(3x-4)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{\cos(3x+3h-4) \cdot \cos(3x-4)} \times \lim_{h \rightarrow 0} \frac{\sin 3h}{h}$$

$$= \frac{1}{\cos(3x-4) \cdot \cos(3x-4)} \times \lim_{h \rightarrow 0} \frac{\sin 3h}{3h} \times 3$$

$$= \frac{1}{\cos^2(3x-4)} \times 1 \times 3$$

$$= 3 \sec^2(3x-4)$$



④ d)  $\cos \frac{3x}{2}$

$\rightarrow$  let  $f(x) = \frac{3x}{2} \cos \frac{3x}{2}$

$f(x+h) = \frac{3(x+h)}{2} \cos \frac{3(x+h)}{2}$

Where  $h$  is small change in  $x$ .

By definition of derivatives,

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{\cos \frac{3(x+h)}{2} - \cos \frac{3x}{2}}{h}$

$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ -2 \sin \left( \frac{3x+3h+3x}{4} \right) \cdot \sin \left( \frac{3x+3h-3x}{4} \right) \right]$

$= \lim_{h \rightarrow 0} -\frac{1}{h} \left[ 2 \sin \left( \frac{6x+3h}{4} \right) \cdot \sin \frac{3h}{4} \right]$

$= \lim_{h \rightarrow 0} -2 \sin \left( \frac{6x+3h}{4} \right) \cdot \frac{\sin \frac{3h}{4}}{\frac{3h}{4}} \times \frac{3}{4}$

$= \lim_{h \rightarrow 0} -2 \sin \left( \frac{6x+3h}{4} \right) \cdot \lim_{h \rightarrow 0} \frac{\sin \frac{3h}{4}}{\frac{3h}{4}} \times \frac{3}{4}$

$= -2 \sin \left( \frac{6x}{4} \right) \cdot 1 \times \frac{3}{4}$

$= -\frac{3}{2} \sin \left( \frac{3x}{2} \right)$



5)  $\sec a\pi$

→ let  $f(x) = \sec a\pi$

$$f(x+h) = \sec a(x+h)$$

Where  $h$  is small change in  $x$ .

By definition of derivatives,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} [\sec(a\pi+h) - \sec a\pi]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{1}{\cos(a\pi+h)} - \frac{1}{\cos a\pi} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{\cos a\pi - \cos(a\pi+h)}{\cos a\pi \cdot \cos(a\pi+h)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ \frac{2 \sin\left(\frac{a\pi+h-a\pi}{2}\right) \cdot \sin\left(\frac{a\pi+h-a\pi}{2}\right)}{\cos a\pi \cdot \cos(a\pi+h)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{a\pi+h-a\pi}{2}\right) \cdot \sin \frac{ah}{2}}{\cos a\pi \cdot \cos(a\pi+h) \cdot \frac{ah}{2} \times \frac{a}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{\sin \frac{ah}{2}}{\frac{ah}{2}} \times \frac{a}{2} \times \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{a\pi+h-a\pi}{2}\right)}{\cos a\pi \cdot \cos(a\pi+h)}$$

$$= \lim_{h \rightarrow 0} 1 \times \frac{a}{2} \times 2 \times \frac{\sin a\pi}{\cos a\pi \cdot \cos a\pi}$$

$$= a \tan a\pi \times \sec a\pi$$



Q6)  $\sin^2 3x$

→ let  $y = \sin^2 3x = f(x)$

$$f(x+h) = \sin^2 3(x+h)$$

$$= \sin^2 (3x+3h)$$

Where  $h$  is small change in  $x$ .

By definition of derivatives,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^2 (3x+3h) - \sin^2 3x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[ [\sin(3x+3h) + \sin 3x] (\sin(3x+3h) - \sin 3x) \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \times 2 \cos \left( \frac{3x+3h+3x}{2} \right) \cdot \sin \left( \frac{3x+3h-3x}{2} \right)$$

$$\times \lim_{h \rightarrow 0} \sin(3x+3h) + \sin 3x$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \times 2 \cos \left( \frac{6x+3h}{2} \right) \cdot \sin \frac{3h}{2} \times (\sin 3x + \sin 3x)$$

$$= \lim_{h \rightarrow 0} \cancel{2} \cos \left( \frac{6x+3h}{2} \right) \cdot \sin \frac{3h}{2} \times \frac{3}{2} \times \cancel{\sin} (2 \sin 3x)$$

$$= \lim_{h \rightarrow 0} \cos \left( \frac{6x+3h}{2} \right) \times 1 \times 3 \times 2 \sin 3x$$

$$= \cos 3x \times 1 \times 3 \times 2 \sin 3x$$

$$= 3 \times 2 \sin 3x \times \cos 3x$$

$$= 3 \times \sin 2 \times 3x$$

$$= 2 \sin 6x$$



7) g)  $\sqrt{\sin 2x}$

→ let  $y = \sqrt{\sin 2x} = f(x)$

$f(x+h) = \sqrt{\sin 2(x+h)}$

Where  $h$  is small change in  $x$ .

By definition of derivatives,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\sin 2(x+h)} - \sqrt{\sin 2x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\sin(2x+2h)} - \sqrt{\sin 2x}}{h} \times \frac{\sqrt{\sin(2x+2h)} + \sqrt{\sin 2x}}{\sqrt{\sin(2x+2h)} + \sqrt{\sin 2x}}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(2x+2h) - \sin 2x}{\sqrt{\sin(2x+2h)} + \sqrt{\sin 2x}} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{2x+2h+2x}{2}\right) \cdot \sin\left(\frac{2x+2h-2x}{2}\right)}{(\sqrt{\sin(2x+2h)} + \sqrt{\sin 2x}) \times h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{4x+2h}{2}\right)}{(\sqrt{\sin(2x+2h)} + \sqrt{\sin 2x})} \times \frac{\sin h}{h}$$

$$= \frac{2 \cos 2x}{\sqrt{\sin 2x} + \sqrt{\sin 2x}} \times \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$= \frac{2 \cos 2x}{2 \sqrt{\sin 2x}} \times 1$$

$$= \frac{\cos 2x}{\sqrt{\sin 2x}}$$

Ans.



8)  $\sqrt{\tan x}$

→ let  $f(x) = \sqrt{\tan x}$

$f(x+h) = \sqrt{\tan(x+h)}$

Where  $h$  is small change in  $x$ .

By definition of derivatives,

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{\sqrt{\tan(x+h)} - \sqrt{\tan x}}{h}$

$= \lim_{h \rightarrow 0} \frac{\sqrt{\tan(x+h)} - \sqrt{\tan x}}{h} \times \frac{\sqrt{\tan(x+h)} + \sqrt{\tan x}}{\sqrt{\tan(x+h)} + \sqrt{\tan x}}$

$= \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{\sqrt{\tan(x+h)} + \sqrt{\tan x}} \times \frac{1}{h}$

$= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{\cos(x+h) \cos x} \times \lim_{h \rightarrow 0} \frac{1}{h(\sqrt{\tan(x+h)} + \sqrt{\tan x})}$

$= \lim_{h \rightarrow 0} \frac{\sin(x+h) \cdot \cos x - \cos(x+h) \cdot \sin x}{\cos x \cdot \cos(x+h)}$

$\times \frac{1}{h(\sqrt{\tan x} + \sqrt{\tan x})}$

$= \lim_{h \rightarrow 0} \frac{\sin(x+h-x)}{\cos x \cdot \cos(x+h)} \times \frac{1}{h(2\sqrt{\tan x})}$

$= \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \lim_{h \rightarrow 0} \frac{1}{\cos x \cdot \cos(x+h)} \times \frac{1}{2\sqrt{\tan x}}$

$= 1 \times \frac{1}{\cos x \cdot \cos x} \times \frac{1}{2\sqrt{\tan x}}$

$= \frac{1}{\cos^2 x} \times \frac{1}{2\sqrt{\tan x}}$

$= \frac{\sec^2 x}{2\sqrt{\tan x}}$



2. Find, the derivative of

9 a)  $\sin 3x$

→ let  $y = \sin 3x$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \sin 3x}{dx}$$

$$= \frac{d \sin 3x}{d 3x} \times \frac{d 3x}{dx}$$

$$= \cos 3x \times 3$$

$$= 3 \cos 3x \quad \downarrow$$

10 b)  $\sin x^3$

→ let  $y = \sin x^3$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \sin x^3}{dx}$$

$$= \frac{d \sin x^3}{d x^3} \times \frac{d x^3}{dx}$$

$$= \cos x^3 \times 3x^2$$

$$= 3x^2 \cos x^3 \quad \downarrow$$

11 c)  $\sin^3 x$

→ let  $y = \sin^3 x$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \sin^3 x}{dx}$$

$$= \frac{d \sin^3 x}{d \sin x} \times \frac{d \sin x}{dx}$$

$$= 3 \sin^2 x \times \cos x$$

$$= 3 \sin^2 x \cdot \cos x \quad \downarrow$$



12) d)  $\sin(4x-5)$

→ let  $y = \sin(4x-5)$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \sin(4x-5)}{dx}$$

$$= \frac{d \sin(4x-5)}{d(4x-5)} \times \frac{d(4x-5)}{dx}$$

$$= \cos(4x-5) \times 4$$

$$= 4 \cos(4x-5) \quad \checkmark$$

13) e)  $\cos(ax^2+b)$

→ let  $y = \cos(ax^2+b)$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \cos(ax^2+b)}{dx}$$

$$= \frac{d \cos(ax^2+b)}{d(ax^2+b)} \times \frac{d(ax^2+b)}{dx}$$

$$= -\sin(ax^2+b) \times 2ax$$

$$= -2ax \sin(ax^2+b) \quad \checkmark$$

14) f)  $\tan \sqrt{x}$

→ let  $y = \tan \sqrt{x}$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \tan \sqrt{x}}{dx}$$

$$= \frac{d \tan \sqrt{x}}{d\sqrt{x}} \times \frac{d\sqrt{x}}{dx}$$

$$= \sec^2 \sqrt{x} \times \frac{1}{2} x^{-1/2}$$

$$= \frac{\sec^2 \sqrt{x}}{2\sqrt{x}} \quad \checkmark$$



15) g)  $\cos(1-x^2)^2$

→ let  $y = \cos(1-x^2)^2$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \cos(1-x^2)^2}{dx}$$

$$= \frac{d \cos(1-x^2)^2}{d(1-x^2)^2} \times \frac{d(1-x^2)^2}{dx}$$

$$= -\sin(1-x^2)^2 \times \frac{d(1-2x^2+x^4)}{dx}$$

$$= -\sin(1-x^2)^2 \times (-4x+4x^3)$$

$$= -\sin(1-x^2)^2 \times -4x(1-x^2)$$

$$= 4x(1-x^2) \cdot \sin(1-x^2)^2$$

16) h)  $\operatorname{Cosec} \frac{1}{x}$

→ let  $y = \operatorname{Cosec} \frac{1}{x}$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \operatorname{Cosec} \frac{1}{x}}{dx}$$

$$= \frac{d \operatorname{Cosec} x^{-1}}{dx^{-1}} \times \frac{dx^{-1}}{dx}$$

$$= -\operatorname{Cosec} x^{-1} \cdot \cot x^{-1} \times (-1x^{-2})$$

$$= + \frac{1}{x^2} \times \operatorname{Cosec} \frac{1}{x} \times \cot \frac{1}{x}$$

$$= \frac{+ \operatorname{Cosec} \frac{1}{x} \times \cot \frac{1}{x}}{x^2}$$



17 i)  $\sec \frac{an^2}{b}$

→ let  $y = \sec \frac{an^2}{b}$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d \sec \frac{an^2}{b}}{dn}$$

$$= \frac{d \sec \frac{an^2}{b}}{d \frac{an^2}{b}} \times \frac{d \frac{an^2}{b}}{dn}$$

$$= \sec \frac{an^2}{b} \times \tan \frac{an^2}{b} \times \left[ b \cdot \frac{d \frac{an^2}{b}}{dn} + \frac{an^2}{b} \cdot \frac{db}{dn} \right]$$

$$= \sec \frac{an^2}{b} \times \tan \frac{an^2}{b} \times b$$

$$= \sec \frac{an^2}{b} \times \tan \frac{an^2}{b} \times \frac{2abn}{b^2}$$

$$= \frac{2an}{b} \sec \frac{an^2}{b} \times \tan \frac{an^2}{b}$$

18 j)  $\sin \left( \frac{1+n^2}{1-n^2} \right) = y$

→ differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d \left[ \sin \left( \frac{1+n^2}{1-n^2} \right) \right]}{dn}$$

$$= \frac{d \left[ \sin \left( \frac{1+n^2}{1-n^2} \right) \right]}{d \left( \frac{1+n^2}{1-n^2} \right)} \times \frac{d \left( \frac{1+n^2}{1-n^2} \right)}{dn}$$

$$= \cos \left( \frac{1+n^2}{1-n^2} \right) \times \frac{(1-n^2) \frac{d(1+n^2)}{dn} - (1+n^2) \frac{d(1-n^2)}{dn}}{(1-n^2)^2}$$



$$= \cos\left(\frac{1+x^2}{1-x^2}\right) \times \frac{(1-x^2)2x + (1+x^2)2x}{(1-x^2)^2}$$

$$= \cos\left(\frac{1+x^2}{1-x^2}\right) \times \frac{[2x - \cancel{2x^3} + 2x + \cancel{2x^3}]}{(1-x^2)^2}$$

$$= \cos\left(\frac{1+x^2}{1-x^2}\right) \times \frac{4x}{(1-x^2)^2}$$

$$= \frac{4x}{(1-x^2)^2} \times \cos\left(\frac{1+x^2}{1-x^2}\right)_{\text{H.}}$$

3. Find the differential coefficients of

19 a)  $\tan(\sin(ax+b))$

→ let  $y = \tan(\sin(ax+b))$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d[\tan(\sin(ax+b))]}{dx}$$

$$= \frac{d \tan(\sin(ax+b))}{d(\sin(ax+b))} \times \frac{d \sin(ax+b)}{d(ax+b)} \times \frac{d(ax+b)}{dx}$$

$$= \sec^2(\sin(ax+b)) \times \cos(ax+b) \times a$$

$$= a \cos(ax+b) \times \sec^2(\sin(ax+b))_{\text{H.}}$$

20 b)  $\sin(\cos\sqrt{x})$

→ let  $y = \sin(\cos\sqrt{x})$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d \sin(\cos\sqrt{x})}{dx}$$

$$= \frac{d \sin(\cos\sqrt{x})}{d(\cos\sqrt{x})} \times \frac{d \cos\sqrt{x}}{d\sqrt{x}} \times \frac{d\sqrt{x}}{dx}$$



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$$= \cos(\cos\sqrt{n}) \times (-\sin\sqrt{n}) \times -\frac{1}{2} n^{-3/2}$$

$$= -\frac{1}{2\sqrt{n^3}} \times \cos(\cos\sqrt{n}) \times \sin\sqrt{n}$$

21 c)  $\tan^5(\cos(Pn-q))$

→ let  $y = \tan^5(\cos(Pn-q))$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d}{dn} \tan^5(\cos(Pn-q))$$

$$= \frac{d \tan^5(\cos(Pn-q))}{d \tan(\cos(Pn-q))} \times \frac{d \tan(\cos(Pn-q))}{d \cos(Pn-q)} \\ \times \frac{d \cos(Pn-q)}{d(Pn-q)} \times \frac{d(Pn-q)}{dn}$$

$$= 5 \tan^4(\cos(Pn-q)) \times \sec^2(\cos(Pn-q)) \\ \times [-\sin(Pn-q)] \times P \\ = -5P \tan^4(\cos(Pn-q)) \cdot \sec^2(\cos(Pn-q)) \\ \sin(Pn-q) \quad \text{Ans.}$$

22 d)  $\sec^3(\tan\sqrt{2n})$

→ let  $y = \sec^3(\tan\sqrt{2n})$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d}{dn} [\sec^3(\tan\sqrt{2n})]$$

$$= \frac{d [\sec^3(\tan\sqrt{2n})]}{d \sec(\tan\sqrt{2n})} \times \frac{d \sec(\tan\sqrt{2n})}{d \tan(\sqrt{2n})} \\ \times \frac{d \tan\sqrt{2n}}{d\sqrt{2n}} \times \frac{d\sqrt{2n}}{dn}$$

$$= 3 \sec^2(\tan\sqrt{2n}) \times \sec(\tan\sqrt{2n}) \cdot \tan(\tan\sqrt{2n}) \\ \times \sec^2\sqrt{2n} \times \frac{1}{\sqrt{2n}}$$



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$$= \frac{3}{\sqrt{2x}} \times \sec^3(\tan\sqrt{2x}) \times \tan(\tan\sqrt{2x}) \times \sec^2\sqrt{2x} \quad \frac{1}{2}$$

4. Find the differential coefficients of

23 a)  $(x^2+3x) \sin 5x$

→ let  $y = (x^2+3x) \sin 5x$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d}{dx} [(x^2+3x) \cdot \sin 5x]$$

$$= \sin 5x \frac{d(x^2+3x)}{dx} + (x^2+3x) \frac{d \sin 5x}{dx}$$

$$= \sin 5x (2x+3) + (x^2+3x) \times \frac{d \sin 5x}{dx} \times \frac{d 5x}{dx}$$

$$= (2x+3) \sin 5x + (x^2+3x) \cos 5x \times 5$$

$$= (2x+3) \sin 5x + 5(x^2+3x) \cos 5x \quad \frac{1}{2}$$

24 b)  $x^3 \tan(2x^2+3x)$

→ let  $y = x^3 \tan(2x^2+3x)$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d}{dx} [x^3 \tan(2x^2+3x)]$$

$$= x^3 \frac{d \tan(2x^2+3x)}{dx} + \tan(2x^2+3x) \times \frac{d x^3}{dx}$$

$$= x^3 \frac{d \tan(2x^2+3x)}{d(2x^2+3x)} \times \frac{d(2x^2+3x)}{dx}$$

$$+ 3x^2 \tan(2x^2+3x)$$

$$= x^3 \sec^2(2x^2+3x) \times (4x+3) + 3x^2 \tan(2x^2+3x)$$

$$= x^3(4x+3) \sec^2(2x^2+3x) + 3x^2 \tan(2x^2+3x) \quad \frac{1}{2}$$



25) c)  $x \cdot \cot^3 x$

→ let  $y = x \cdot \cot^3 x$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d[x \cdot \cot^3 x]}{dx}$$

$$= x \frac{d \cot^3 x}{dx} + \cot^3 x \frac{dx}{dx}$$

$$= x \frac{d \cot^3 x}{d \cot x} \times \frac{d \cot x}{dx} + \cot^3 x$$

$$= 3 \cot^2 x \times (-\operatorname{cosec}^2 x) + \cot^3 x$$

$$= -3 \cot^2 x \times \operatorname{cosec}^2 x + \cot^3 x$$

$$= -\cot^2 x (3 \operatorname{cosec}^2 x + \cot x) \quad \text{Ans.}$$

26) d)  $\frac{1}{\sqrt{x}} \cos \sqrt{x}$

→ let  $y = \frac{\cos \sqrt{x}}{\sqrt{x}}$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d\left[\frac{\cos \sqrt{x}}{\sqrt{x}}\right]}{dx}$$

$$= \frac{\sqrt{x} \frac{d \cos \sqrt{x}}{d \sqrt{x}} - \cos \sqrt{x} \frac{d \sqrt{x}}{dx}}{(\sqrt{x})^2}$$

$$= \left( \sqrt{x} \cdot \frac{d \cos \sqrt{x}}{d \sqrt{x}} \times \frac{d \sqrt{x}}{dx} - \cos \sqrt{x} \times \frac{1}{2} x^{-1/2} \right) \times \frac{1}{x}$$

$$= \left( \sqrt{x} \cdot \frac{d \cos \sqrt{x}}{d \sqrt{x}} \times \frac{1}{2\sqrt{x}} - \cos \sqrt{x} \times \frac{1}{2\sqrt{x}} \right) \times \frac{1}{x}$$

$$= \left( \sqrt{x} \cdot (-\sin \sqrt{x}) \times \frac{1}{2\sqrt{x}} - \cos \sqrt{x} \times \frac{1}{2\sqrt{x}} \right) \times \frac{1}{x}$$

$$= \left( -\frac{1}{2} \sin \sqrt{x} - \frac{\cos \sqrt{x}}{2\sqrt{x}} \right) \times \frac{1}{x}$$



$$= -\frac{1}{2n} \left[ \sin\sqrt{n} + \frac{\cos\sqrt{n}}{\sqrt{n}} \right] \#$$

27 e)  $\tan n^2 \cdot \sin^2 n$

→ let  $y = \tan n^2 \cdot \sin^2 n$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d[\tan n^2 \cdot \sin^2 n]}{dn}$$

$$= \sin^2 n \frac{d \tan n^2}{dn} + \tan n^2 \frac{d \sin^2 n}{dn}$$

$$= \sin^2 n \times \frac{d \tan n^2}{dn^2} \times \frac{dn^2}{dn} + \tan n^2 \times \frac{d \sin^2 n}{d \sin n} \times \frac{d \sin n}{dn}$$

$$= \sin^2 n \times \sec n^2 \times 2n + \tan n^2 \times 2 \sin n \times \cos n$$

$$= 2n \sin^2 n \cdot \sec n^2 + 2 \sin 2n \cdot \tan n^2$$

28 f)  $\frac{\sec n}{2n-3}$

→ let  $y = \frac{\sec n}{2n-3}$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d(\sec n)}{(2n-3)}$$

$$= \frac{(2n-3) \frac{d \sec n}{dn} - \sec n \frac{d(2n-3)}{dn}}{(2n-3)^2}$$

$$= \frac{(2n-3) \times \frac{d \sec n}{dn} \times \frac{dn}{dn} - \sec n \times 2}{(2n-3)^2}$$

$$= \frac{1}{(2n-3)^2} [a(2n-3) \cdot \sec n \cdot \tan n - 2 \sec n]$$



$$= \frac{\sec n [a(2n-3) \tan n - 2]}{(2n-3)^2}$$

5. Find the derivative of the following.

29 a)  $\frac{1}{\sec n - \tan n}$

→ let  $y = \frac{1}{\sec n - \tan n}$

differentiating with respect to  $n$ .

$$\frac{dy}{dn} = \frac{d}{dn} \left[ \frac{1}{\sec n - \tan n} \right]$$

$$= \sec n - \tan n \frac{d1}{dn} - 1 \frac{d(\sec n - \tan n)}{dn}$$

$$= - \left[ \frac{d \sec n}{dn} - \frac{d \tan n}{dn} \right] \times \frac{1}{(\sec n - \tan n)^2}$$

$$= - \frac{1}{(\sec n - \tan n)^2} \times [\sec n \cdot \tan n - \sec^2 n]$$

$$= \frac{\sec^2 n - \sec n \cdot \tan n}{(\sec n - \tan n)^2}$$

$$= \frac{\sec n (\sec n - \tan n)}{(\sec n - \tan n)^2}$$

$$= \frac{\sec n}{\sec n - \tan n} \times \frac{\sec n + \tan n}{\sec n + \tan n}$$

$$= \frac{\sec n (\sec n + \tan n)}{\sec^2 n - \tan^2 n}$$

$$= \sec n (\sec n + \tan n)$$



30 b)  $\frac{1 - \cos 2x}{\sin 2x}$

→ let  $y = \frac{1 - \cos 2x}{\sin 2x}$

differentiating with respect to  $x$ .

$$\frac{dy}{dx} = \frac{d}{dx} \left( \frac{1 - \cos 2x}{\sin 2x} \right)$$

$$= \frac{\sin 2x \frac{d(1 - \cos 2x)}{dx} - (1 - \cos 2x) \frac{d \sin 2x}{dx}}{\sin^2 2x}$$

$$= \frac{1}{\sin^2 2x} \left[ \sin 2x \left( \frac{d(-\cos 2x)}{dx} \right) \times \frac{dx}{dx} - (1 - \cos 2x) \frac{d \sin 2x}{dx} \times \frac{dx}{dx} \right]$$

$$= \frac{1}{\sin^2 2x} \left[ \sin 2x \cdot \sin 2x \cdot 2 - (1 - \cos 2x) \cdot \cos 2x \times 2 \right]$$

$$= \frac{1}{\sin^2 2x} [2 \sin^2 2x - 2 \cos 2x (1 - \cos 2x)]$$

$$= \frac{1}{\sin^2 2x} [2 \sin^2 2x - 2 \cos 2x + 2 \cos^2 2x]$$

$$= \frac{1}{\sin^2 2x} [2(\sin^2 2x + \cos^2 2x) - 2 \cos 2x]$$

$$= \frac{1}{\sin^2 2x} [2 - 2 \cos 2x]$$

$$= \frac{2(1 - \cos 2x)}{\sin 2x \cdot \sin 2x}$$



$$= \frac{2(1 - \cos 2x)}{2 \sin x \cdot \cos x \cdot \sin 2x}$$

$$= \frac{x - x + 2 \sin^2 x}{\sin x \cdot \cos x \cdot \sin 2x}$$

$$= \frac{2 \sin x \cdot \sin x}{\sin x \cdot \cos x \cdot 2 \sin x \cdot \cos x}$$

$$= \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$

$$\Rightarrow \frac{\frac{1}{\cos x}}{\frac{1}{\cos x}}$$

$$\rightarrow \text{let } y = \frac{1 - \cos x}{1 + \cos x}$$

$$= \frac{(1 - \cos x) \times (1 - \cos x)}{(1 + \cos x)(1 - \cos x)}$$

$$= \frac{(1 - \cos x)^2}{1 - \cos^2 x}$$

$$= \frac{(1 - \cos x)^2}{\sin^2 x}$$

~~differentiating~~ with respect to  $x$ .

$$\frac{d}{dx} \left[ \frac{(1 - \cos x)^2}{\sin^2 x} \right]$$

$$= \frac{\sin^2 x \frac{d}{dx} (1 - \cos x)^2 - (1 - \cos x)^2 \frac{d}{dx} \sin^2 x}{\sin^4 x}$$



31 c)  $\frac{1 - \cos n}{1 + \cos n}$

→ let  $y = \frac{1 - \cos n}{1 + \cos n}$   
 $= \frac{2 \sin^2 n/2}{2 \cos^2 n/2}$

$y = \tan^2 n/2$

differentiating with respect to  $n$ .

$\frac{dy}{dn} = \frac{d \tan^2 n/2}{dn}$

$= \frac{d \tan^2 n/2}{d \tan n/2} \times \frac{d \tan n/2}{d n/2} \times \frac{d n/2}{dn}$

$= 2 \tan \frac{n}{2} \times \sec^2 \frac{n}{2} \times \frac{1}{2}$

$= \tan \frac{n}{2} \times \sec^2 \frac{n}{2}$

32 d)  $\frac{\cos 2n}{1 - \sin 2n}$

→ let  $y = \frac{\cos 2n}{1 - \sin 2n}$

$= \frac{\cos^2 n - \sin^2 n}{\cos^2 n + \sin^2 n - 2 \sin n \cdot \cos n}$

$= \frac{(\cos n + \sin n)(\cos n - \sin n)}{(\cos n - \sin n)^2}$

$= \frac{\cos n + \sin n}{\cos n - \sin n}$

differentiating with respect to  $n$ .

$\frac{dy}{dn} = \frac{d}{dn} \left[ \frac{\cos n + \sin n}{\cos n - \sin n} \right]$